Problem set 1

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Lecture 3

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Contents

1 Problem 1 **2**

2 Problem 2 **3**

3 Problem 3 **4**

4 Problem 4 **5**

5 Problem 5 **6**

6 Problem 6 **7**

7 Problem 7 **8**

8 Problem 8 **9**

1 Problem 1

A product is produced at three factories (A, B, C) and is shipped to four markets (1, 2, 3, 4)

Capacities at the three factories:

A	**B**	**C**
85	90	110

Demand at the four markets:

1	**2**	**3**	**4**
55	60	35	50

Variable cost per unit for production and shipping:

	To 1	**To 2**	**To 3**	**To 4**
From A	16	13	18	24
From B	21	26	23	27
From C	24	23	22	29

Formulate a mathematical model to minimize total costs

Variables:

- X_{ij}: amount of product shipped from factory i to market j

Parameters:

- d_j: demand at market j
- cap_i: capacity at factory i
- c_{ij}: cost for production at factory i plus shipping to market j

Model:

$$\min \sum_i \sum_j X_{ij} c_{ij}$$

$$\text{s.t.}$$

$$\sum_i X_{ij} = d_j \quad \forall j$$

$$\sum_j X_{ij} \leq \text{cap}_i \quad \forall i$$

$$X_{ij} \geq 0 \quad \forall i, j$$

Optimal shipments:

From/To	**Amount**
A $\rightarrow$ 1	25
A $\rightarrow$ 2	60
B $\rightarrow$ 1	30
B $\rightarrow$ 4	50
C $\rightarrow$ 3	35

Total cost = 3930

2 Problem 2

The pyrotec company produces three electrical products; clocks, radios and toasters. These products have the following resource requirements:

	Cost/unit	**labor hours / unit**
Clock	7	2
Radio	10	3
Toaster	5	2

The manufacturer has a daily production budget of 2,000 and a maximum of 660 hours of labor. Maximum daily customer demand is for 200 clocks, 300 radios and 150 toasters. Clocks sell for 15, radios for 20 and toasters for 12. The company wants to know the optimal product mix that will maximise profit

Decision variables:

- X_i number of product i produced and sold

Parameters:

- c_i: cost per unit i
- l_i: labor hours per unit for product i
- b: daily production budget
- cap: maximum hours of labor
- d_i: demand for product i
- p_i: price per unit i

Model:

$$\max \sum_i X_i(p_i - c_i)$$
$$\text{s.t.}$$
$$0 \leq X_i \leq d_i \quad \forall i$$
$$\sum_i X_i c_i \leq b$$
$$\sum_i X_i l_i \leq \text{cap}$$

Optimal production:

Product	**Amount**
Clock	177
Radio	1
Toaster	150

Total profit: 2476

(note: solution is a float, but that is technically possible since we cannot sell a third of a toaster)

3 Problem 3

Consider the following transportation problem:

From	**Cost to A**	**Cost to B**	**Cost to C**	**Supply**
A	6	9	100	130
B	12	3	5	70
C	4	8	11	100
Demand	80	110	60	

I guess we need to maximize the profit while satisfying the supply

Model:

$$\max \sum_i (p_i - c_i) X_i$$

I do not fully understand the objective of this problem so I will not participate

4 Problem 4

Solve the following problem using a set-based PuLP model:

$$\min Z = 3x_{11} + 12x_{12} + 8x_{13} + 10x_{21} + 5x_{22} + 6x_{23} + 6x_{31} + 7x_{32} + 10x_{33}$$

$$\text{s.t.}$$

$$x_{11} + x_{12} + x_{13} = 90$$
$$x_{21} + x_{22} + x_{23} = 30$$
$$x_{31} + x_{32} + x_{33} = 100$$
$$x_{11} + x_{21} + x_{31} \leq 70$$
$$x_{12} + x_{22} + x_{32} \leq 110$$
$$x_{13} + x_{23} + x_{33} \leq 80$$
$$x_{ij} \geq 0$$

Optimal shipments:

From/To	**Amount**
x_{11}	70
x_{12}	0
x_{13}	20
x_{21}	0
x_{22}	10
x_{23}	20
x_{31}	0
x_{32}	100
x_{33}	0

Minimum cost = 1240.0

5 Problem 5

A hospital dietitian prepares breakfast menus every morning for the hospital patients. Part of the dietitian's responsibility is to make sure that minimum daily requirements for vitamin A and B are met. At the same time, the cost of the menus must be kept as low as possible. The main breakfast staples providing vitamins A and B are eggs, bacon and cereal. The vitamin requirements and vitamin contributions for each staple follow:

Vitamin	egg	bacon	cereal	minimum daily req.
A	2	4	1	16
B	3	2	1	12

An egg costs 0.04, a bacon strip costs 0.03, and a cereal costs 0.02. The dietitian wants to know how much of each staple to serve per order to meet the minimum daily vitamin requirements while minimizing total cost.

Decision Variable:

- X_i: Amount of food type i

Parameters:

- c_i: cost per food i
- R_A: daily requirement of vitamin A
- R_B: daily requirement of vitamin B
- A_i: vitamin A in food i
- B_i: vitamin B in food i

Model:

$$\min \sum_i X_i c_i$$

$$\text{s.t.}$$

$$\sum_i A_i X_i \geq R_A$$

$$\sum_i B_i X_i \geq R_B$$

$$X_i \geq 0 \quad \forall i$$

Food	**Amount**
Egg	2
Bacon	3
Cereal	0

Total cost: 0.17

6 Problem 6

Suppose the foods listed below have calories, protein, calcium, vitamin A, and costs per pound as shown. In what amounts should these foods be purchased to meet at least the daily requirements while minimizing the total costs?

	bread	**meat**	**potatoes**	**cabbage**	**milk**	**gelatin**	**required**
calories	1254	1457	318	46	309	1725	3000
protein	39	73	8	4	15	45	70 g
calcium	418	41	42	141	536	0	800 mg
vitamin A	0	0	70	860	720	0	500 IU
cost/pound	0.3	1	0.05	0.08	0.23	0.48	

Decision variable:

- X_i: amount of food i (lbs)

Parameters:

- c_i: cost / lb for food i
- R_j: required micro/macro nutrient j
- F_{ij}: amount of micro/macro nutrient j in food i

Model:

$$\min \sum_i c_i X_i$$

$$\text{s.t.}$$

$$\sum_i F_{ij} X_i \geq R_j \quad \forall j$$

$$X_i \geq 0 \quad \forall i$$

7 Problem 7

Bayville has built a new elementary school, increasing the town's total to four schools — Addison, Beeks, Canfield and Daley. Each has a capacity of 400 students. The school board wants to assign children to schools s.t. their travel time by bus is as short as possible. The school board has partitioned the town into five districts conforming to population density — north, south, east, west and central. The average bus travel time from each district to each school is shown as follows:

District	Addison	Beeks	Canfield	Daley	**Student population**
North	12	23	35	17	250
South	26	15	21	27	340
East	18	20	22	31	310
West	29	24	35	10	210
Central	15	10	23	16	290

Determine the number of children should be assigned from each district to each school to minimize total student travel time

Decision variable:

- X_{ij}: number of students from district j assigned to school i

Parameters:

- t_{ij}: travel time from district j to school i
- P_j: student population in district j
- c_i: capacity for school i (400 for all of the schools)

Model:

$$\min \sum_i \sum_j X_{ij} t_{ij}$$

$$\text{s.t.}$$

$$\sum_i X_{ij} = P_j \quad \forall j$$

$$\sum_j X_{ij} \leq c_i \quad \forall i$$

$$X_{ij} \geq 0 \quad \forall i, j$$

8 Problem 8

World Food, Inc., imports food products such as meats, cheeses and pastries